

A two-stage energy management strategy for household microgrids considering the influence of electric vehicle travel characteristics and daylight hours

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Abstract

We propose a two-stage energy management strategy for household microgrids to improve their operational efficiency and economic benefits by making full use of the time-of-use electricity price and considering the impact of temporal and spatial uncertainty of electric vehicles (EVs) and different sunshine hours on household photovoltaic (PV). The strategy combines the time-sharing tariff period and the travel time of EVs to make full use of the dual attributes of EVs as both load and generation units. Particle swarm algorithm is used to optimize the household PV output, EV and storage system charging/discharging strategies, and the energy interaction between the microgrid and the main network. This algorithm guides the household microgrid to achieve the optimal economic operation while cutting the peaks and filling the valleys of the main network to realize the efficient allocation of power resources. Finally, three typical scenarios are used for optimization, and the results show that the proposed energy management strategy can provide savings of 1,413.02, 1,026.94 and 929.19 yuan per day under the three scenarios analyzed, respectively, which verifies the effectiveness and rationality of the model. A sensitivity analysis is also conducted.

KEYWORDS

electric vehicle, household microgrid, TOU electricity price, two-stage energy management strategy

1 | INTRODUCTION

Global warming is one of the biggest challenges facing mankind today. China, as the country with the largest total and incremental carbon emissions in the world, put forward the carbon peak and carbon neutral target in September 2020. In June 2021, the National Energy Administration (NEA) issued a policy to promote

distributed photovoltaic (PV) in whole counties and vigorously support household PV projects, making household PV one of the most promising forms of residential energy use.

However, household PV output is affected by seasonal and weather conditions,¹ and the output power is volatile and random, which has a negative impact on grid security and stability and power quality.^{2,3} In addition, PV

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cannot adjust the output power and cannot generate power at any time to meet the load demand.⁴ Therefore, the introduction of electric vehicles (EVs) with a large number of idle power batteries⁵ is an effective way to regulate household PV output. Through technologies such as vehicle to grid (V2G) and vehicle to home (V2H),⁶ EVs can not only effectively smooth PV output power by using their own energy transfer capability, but also achieve peak and valley reduction.⁷ How to effectively regulate EVs has thus become a current research hotspot. In the study by Yangyang et al.,⁸ based on the EV charging network planning problem, an opportunity-constrained planning model that also considers the stochastic characteristics of distributed PV output EV charging load is established. Zhiwei et al.⁹ and Clairand et al.¹⁰ introduced agents to participate in EV scheduling, integrate EV charging and discharging power, and participate in energy management.

Although scholars have carried out in-depth research on EV control strategies, EVs should always focus on satisfying users' travel needs¹¹ and are not flexible enough to regulate household PV output, therefore need traditional energy storage systems (ESS) to coordinate and work together to form a household microgrid¹² to accomplish more effective and deeper dispatch. Jingli et al.¹³ combined EVs, energy storage devices, and PV systems to study the configuration optimization problem in the process of accessing the distribution network and proposed an optimization strategy of energy storage capacity considering the orderly charging of EVs. Su et al.¹⁴ proposed an energy management strategy that takes into account PV output, EV charging and discharging power, and energy storage energy state under time-sharing tariff based on the power calculation model of EV battery charge state energy management.

Although the aforementioned scholars have conducted a lot of studies on energy management of household microgrid, neither have fully utilized the coupling between time period of time of use (TOU) electricity price and EV travel time, nor have considered the impact of different sunshine hours due to seasonal climate change on the optimized PV output of EVs. Therefore, based on the existing research, this paper proposes a two-stage energy management strategy for home microgrid by taking into account the influence of seasonal weather on PV and the spatiotemporal characteristics of EVs, making full use of the TOU electricity price mechanism, and aiming at the lowest operating cost. Finally, the particle swarm algorithm is used to analyze three typical scenarios to achieve the optimal economic operation of the household microgrid under different scenarios.

Compared with existing studies, the innovation of this paper lies in that (a) for the spatial and temporal

TABLE 1 Typical daily load data

Period	Load (kW)	Period	Load (kW)	Period	Load (kW)
1	54.31	9	92.63	17	89.84
2	50.72	10	96.52	18	91.53
3	48.03	11	96.42	19	90.94
4	46.23	12	94.83	20	84.05
5	43.93	13	84.65	21	77.16
6	46.93	14	78.36	22	58.30
7	58.20	15	79.56	23	56.81
8	71.98	16	81.95	24	55.71

uncertainty of EV charging and discharging power, a Markov chain theory-based model of EV travel and battery power change is established to constrain EV participation in home micro-grid energy management behavior. (b) Making full use of the feature that the TOU period fits with the travel time of EVs, a two-stage energy management strategy for household microgrid under the time-sharing tariff mechanism is constructed with the time-sharing tariff valley period as the first stage and the time-sharing tariff other periods as the second stage. (c) The differentiated energy management strategy is designed by considering the difference of sunshine time caused by seasonal changes and is verified by three typical scenarios (Table 1).

The rest of the paper is organized as follows: Section 2 describes the Markov chain-based uncertainty model for EVs; Section 3 develops a model of a household micro-grid system containing EVs; Section 4 designs a two-stage energy management strategy for household microgrids under time-sharing tariffs; Section 5 presents simulation results to verify the validity of the model; and Section 6 highlights the main conclusions of the paper.

2 | MARKOV CHAIN-BASED MODELING OF UNCERTAINTY BEHAVIOR OF EVs

2.1 | EV travel characteristics volume

EVs can exchange energy through charging and discharging, but the primary goal is to meet the user's travel demand. Therefore, the charging and discharging power of EVs is uncertain in time and space, and the time of grid entry and power consumption depend on the travel habits and driving characteristics of users. In this paper, we simulate EV travel patterns based on the U.S. Department of Transportation survey data¹⁵

(National Household Travel Survey [NHTS]). According to the data processing method proposed by Liting et al.,¹⁶ the fitted EV travel characteristic quantities satisfy the following probability density functions, respectively.

2.1.1 | Start traveling time of the day T_b

$$f_b(x) = \begin{cases} \frac{1}{\sigma_b \sqrt{2\pi}} \exp \left[-\frac{(x - \mu_b)^2}{2\sigma_b^2} \right] & 0 < x \leq (\mu_b + 12), \\ \frac{1}{\sigma_b \sqrt{2\pi}} \exp \left[-\frac{(x - 24 - \mu_b)^2}{2\sigma_b^2} \right] & (\mu_b + 12) < x \leq 24, \end{cases} \quad (1)$$

where μ_b and σ_b are the mean and the variance of the travel time sample data, respectively, and can be taken as 7.84 and 3.16.

2.1.2 | Return time of the day T_e

$$f_e(x) = \begin{cases} \frac{1}{\sigma_e \sqrt{2\pi}} \exp \left[-\frac{(x - \mu_e)^2}{2\sigma_e^2} \right] & (\mu_e - 12) < x \leq 24, \\ \frac{1}{\sigma_e \sqrt{2\pi}} \exp \left[-\frac{(x + 24 - \mu_e)^2}{2\sigma_e^2} \right] & 0 < x \leq (\mu_e - 12), \end{cases} \quad (2)$$

where μ_e and σ_e are the mean and the variance of the return time sample data, preferably 17.8 and 2.94, respectively.

2.1.3 | Daily mileage S_{EV}

$$f_s(x) = \frac{1}{x} \frac{1}{\sigma_s \sqrt{2\pi}} \exp \left[-\frac{(\ln x - \mu_s)^2}{2\sigma_s^2} \right], \quad (3)$$

where μ_s and σ_s are the mean and the variance of the daily mileage sample data and can be taken as 3.5 and 0.85, respectively.

2.2 | Markov chain trip description

Markov chain is a model describing a stochastic dynamic system, which means that the state of the system at each moment is random, the transfer of the state is probabilistic, and the future state distribution is only related to the present state and its transfer probability, but not to the past, that is, there is no posteriority. Let the state at the

current moment be A_i , and the state at the next moment be A_j , then the Markov chain can be expressed in terms of conditional probabilities as:

$$P(A_i \rightarrow A_j) = P(A_j | A_i) = P_{ij}. \quad (4)$$

If the location of the vehicle is considered as a state, it is known from Markov theory that the location of the vehicle in the next hour can be deduced from the current state. Considering the actual activities of EV users, the transfer probability changes all the time in 24 hours a day, for example, the transfer probability of going home in the evening is necessarily higher than that in the early morning, so the transfer matrix should also have the temporal property of the current state. Denote $P_{ij,\tau}$ as the transfer probability from state A_i to state A_j in the time period of τ , then the transfer probability matrix in the time period of τ is:

$$P_\tau = \begin{pmatrix} P_{11,\tau} & \cdots & P_{1n,\tau} \\ \vdots & \ddots & \vdots \\ P_{n1,\tau} & \cdots & P_{nn,\tau} \end{pmatrix}. \quad (5)$$

In the time period of τ , the vehicle is in state A_i , and the next time period may shift to any state $A_j (j \in [1, n])$, and the time period of τ can be any time of the day, so the above transfer matrix should satisfy:

$$\begin{cases} 0 \leq P_{ij,\tau} \leq 1, i, j = 1, 2, \dots, n \\ \sum_j P_{ij,\tau} = 1 \\ 0 \leq \tau \leq 24 \end{cases} \quad (6)$$

Since the EV travel destinations are mainly residential areas (denoted as R), work areas (denoted as W), and other areas (denoted as O), n is 3, which corresponds to state A1, A2, and A3 in turn, the probability matrix of the vehicle moving from one location to another in the time period of τ is shown in Equation 7.

$$\begin{matrix} & \begin{matrix} R & W & O \end{matrix} \\ \begin{matrix} R \\ W \\ O \end{matrix} & \begin{pmatrix} P_{11,\tau} & P_{12,\tau} & P_{13,\tau} \\ P_{21,\tau} & P_{22,\tau} & P_{23,\tau} \\ P_{31,\tau} & P_{32,\tau} & P_{33,\tau} \end{pmatrix} \end{matrix} \quad (7)$$

Based on the vehicle data provided by NHTS, the frequency of vehicle transfers in each time period with different regions as the starting point was counted to obtain the transfer frequency matrix of EVs moving between different regions in

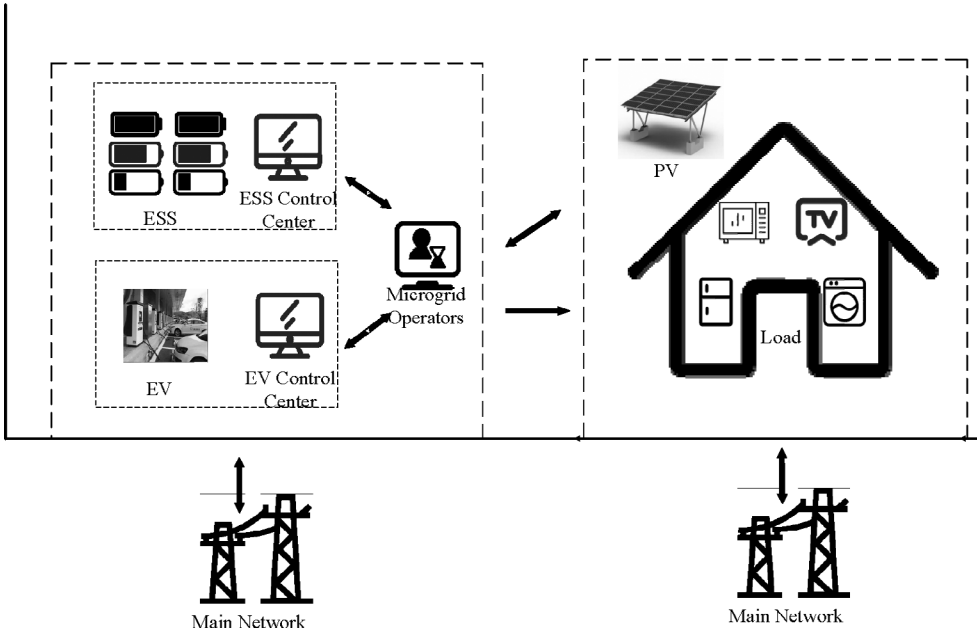


FIGURE 1 Household microgrid structure

each time period. Due to the large amount of NHTS statistics, it is considered that this frequency can be used to represent the spatial transfer probability of EV trips.¹⁷

3 | MODEL OF HOUSEHOLD MICROGRID SYSTEM WITH EVs

The structure of home microgrid system is shown in Figure 1, which mainly consists of household PV, ESS and EV, and is connected to the grid through a contact line to realize the energy interaction (EI) with the main grid. Among them, the services of energy storage and EVs can be provided by the home micro-grid operator. According to the microgrid user load demand, centralized energy storage devices and intelligent charging and discharging control systems are constructed, operated, and maintained to orderly arrange the charging and discharging schedule for all users within the microgrid. When the PV output power is greater than the load demand, the excess power will be stored in ESS and EV, and when the PV output power cannot meet the load demand, the ESS and EV will be called to release power successively, integrating all the energy in the microgrid to participate in the system EI and maximizing the system PV consumption ratio.

3.1 | PV model

Standard test condition (STC) is defined as a condition with a light intensity of 1000 W/m² and a temperature of 25°C. The output power of the PV module can be calculated based on the output power under STC, sunlight

intensity and ambient temperature,^{18,19} which can be defined as

$$P_{PV,\tau} = \frac{P_{STC} G_{C,\tau}}{G_{STC}} [1 + H(T_{C,\tau} - T_{STC})], \quad (8)$$

where $P_{PV,\tau}$ is the output power of household PV in the time period of τ ; P_{STC} and G_{STC} are the maximum test power and the light intensity at STC (1000 W/m²), respectively; $G_{C,\tau}$ is the solar radiation intensity in the time period of τ ; H is the power temperature coefficient, taken as $-0.0047/^{\circ}\text{C}$; $T_{C,\tau}$ is the working temperature of PV panel in the time period of τ ; T_{STC} is the standard test temperature, that is, 25°C.

3.2 | ESS model

The energy storage device can realize the energy transfer of the home microgrid system through the storage and release of energy, optimize the output curve of PV power supply, and ensure the economic operation of the microgrid. Under the current energy storage technology conditions, taking into account the cost performance and other factors, the microgrid energy storage devices are still mainly battery with the charge state of²⁰:

$$\begin{cases} \text{SOC}_{ESS,\tau} = \text{SOC}_{ESS,\tau-1} - \frac{Q_{ESS,\tau}}{\eta_{ESS,out} \times W_{ESS}} & Q_{ESS,\tau} \geq 0, \\ \text{SOC}_{ESS,\tau} = \text{SOC}_{ESS,\tau-1} - \frac{Q_{ESS,\tau} \times \eta_{ESS,in}}{W_{ESS}} & Q_{ESS,\tau} < 0, \end{cases} \quad (9)$$

where $SOC_{ESS,\tau}$ is the charge state of the energy storage battery in the time period of τ ; $Q_{ESS,\tau}$ is the charge and discharge power of the ESS in the time period of τ , positive means the battery is discharged and negative means the battery is charged; $\eta_{ESS,in}$, $\eta_{ESS,out}$ is the charge and discharge efficiency of the energy storage battery respectively; W_{ESS} is the capacity of the energy storage battery.

The charging and discharging constraints of the energy storage device and the power constraints are

$$\begin{cases} SOC_{ESS,min} \leq SOC_{ESS,\tau} \leq SOC_{ESS,max}, \\ -0.2W_{ESS} \leq Q_{ESS,\tau} \leq 0.2W_{ESS}, \end{cases} \quad (10)$$

where SOC_{min} is the lower limit of the state of charge, preferably 0.2; SOC_{max} is the upper limit of the state of charge, preferably 0.9.

The operating cost of ESS can be

$$C_{ESS} = \sum_{\tau=1}^{24} \mu_{ESS} |Q_{ESS,\tau}|, \quad (11)$$

where C_{ESS} is the depreciation loss cost generated by battery energy storage's charging and discharging; μ_{ESS} is the battery loss coefficient.

3.3 | EV model

In this paper, we consider all EVs in a household microgrid as a whole, and study the charging and discharging behavior of a large number of EVs after they are connected, and make the following assumptions.

1. All users in the microgrid are rational users, who can control the charging and discharging behavior of EVs under the guidance of time-sharing tariff and conduct charging and discharging in an orderly manner.
2. Due to the imperfect construction of charging piles in the microgrid, EVs can only be charged and discharged in residential areas.
3. In general, because of the short daily driving mileage of EVs, the amount of discharging in a day is small compared with the total battery capacity, so only a single continuous charging is required in the early morning valley hours, and the charging method is only considered for regular charging.

Then, the daily energy consumption of EV can be calculated from the daily mileage, and the grid charge status can be obtained:

$$SOC_{EV,0} = 1 - \frac{S_{EV}W_{\lambda} + Q_{EV,D}}{W_{EV}}, \quad (12)$$

where $SOC_{EV,0}$ is the charge state of the EV at the time of grid entry; W_{λ} is the power consumption per kilometer of the EV; $Q_{EV,D}$ is the discharge of the EV between the time of travel and the time of return; W_{EV} is the EV battery capacity.

The charge state and power constraints for each time period after the EV is connected to the grid satisfy Equations (9) and (10), and according to the above assumptions there are also constraints:

$$\begin{cases} Q_{EV,\tau} < 0 & 0 < \tau \leq \tau_T, \\ 0 \leq Q_{EV,\tau} \leq 0.2W_{EV} \times D_{EV,\tau} & \tau_T < \tau \leq 24, \end{cases} \quad (13)$$

where W_{EV} is the EV battery capacity; $D_{EV,\tau}$ is the number of dispatchable EVs in the time period of τ (the number of EVs in the residential area); τ_T is the EV travel time.

The depreciation cost resulting from the loss of EV power batteries satisfies Equation (11) and can be expressed as:

$$C_{EV} = \sum_{\tau=1}^{24} \mu_{EV} |Q_{EV,\tau}|. \quad (14)$$

4 | TWO-STAGE ENERGY MANAGEMENT STRATEGY FOR HOUSEHOLD MICROGRID UNDER TOU ELECTRICITY PRICE

Under the TOU electricity price mechanism, to achieve the optimal economic operation of the household microgrid, the peak-valley tariff difference should be fully utilized to guide the ESS and EV to charge during the valley hours and discharge during the other hours. However, since the internal power source of the household microgrid is only PV, the system is strongly affected by the variation of sunshine, and the difference of sunshine time in different seasons will affect the PV output and thus the valley hour ESS and EV operation strategy. Therefore, a two-stage energy management strategy is designed to ensure system economics.

4.1 | Energy management strategy in the first stage

The first stage is the charging stage of ESS and EV during the valley hours of the TOU. Since the feed-in tariff of

home microgrid is higher than the time-sharing valley tariff, the system energy management strategy in this phase is simple, and the goal is to maximize the battery charge state by using the low tariff in the valley. Considering the transport properties of EVs and for prolonging the service life, the charging method is conventional slow charging with grid power purchase, and the charging time is about 5 hours, and the charging power can be expressed as

$$P_{EV,v} = \frac{0.9W_{EV} - W_{EV,R}^{d-1}}{\eta_{EV,in} \times 5}, \quad (15)$$

where $P_{EV,v}$ is the first-stage EV charging power; $W_{EV,R}^{d-1}$ is the remaining EV battery capacity on the previous dispatch date.

ESS is mainly used for home microgrid energy transfer, which can store PV residual power during the charging process, but it is affected by sunrise time, and the

energy management strategy differs in different seasons. As shown in Figure 2, in summer, due to the early sunrise time, the microgrid is still in the valley time, but the sunshine intensity has reached a high level, at this time, the PV output is greater than the load demand, so the residual power can be stored in ESS; while in winter, the sunrise time is late, and the sunshine intensity is always at a low level in the valley time, so the PV output cannot meet the load demand, and ESS can only purchase electricity from the grid and store electricity. The specific optimal management strategies of ESS for different sunrise scenarios are described as follows.

1. The early sunrise condition in summer: During the time period before sunrise, the microgrid operator purchases power from the grid to meet customer loads while simultaneously charging the ESS at constant power:

$$P_{ESS,v}^1 = \frac{0.9W_{ESS} - W_{ESS,R}^{d-1} - \sum_{\tau=\tau_r}^{\tau_v^{\text{end}}-1} P_{PV,\tau} \times 1}{\eta_{ESS,in} \times T_v}, \quad (16)$$

where $P_{ESS,v}^1$ is the ESS charging power before sunrise in summer; $W_{ESS,R}^{d-1}$ is the remaining capacity of the ESS battery on the previous scheduling day; τ_r is the sunrise moment in summer; τ_v^{end} is the end moment of the valley period; T_v is the duration of the valley period.

In the time period after sunrise, when the PV output is less than the load demand, the operator's behavior is the same as before sunrise. When the PV output is greater than the load demand, the remaining power is used for ESS charging, and if the output power is greater than the upper limit of battery charging power, the battery is charged at maximum power.

$$P_{ESS,v}^2 = \begin{cases} P_{ESS,v}^1 + P_{PV,\tau} - P_{U,\tau} & P_{ESS,v}^2 < 0.2W_{ESS}, \\ 0.2W_{ESS} & P_{ESS,v}^2 \geq 0.2W_{ESS}, \end{cases} \quad (17)$$

where $P_{ESS,v}^2$ is the ESS charging power after sunrise in summer; $P_{U,\tau}$ is the power used by customers in the time period of τ .

2. The late sunrise condition in winter: Microgrid operators always purchase power from the grid to meet customer loads while charging ESS and EVs at constant power.

$$P_{ESS,v}^3 = \frac{0.9W_{ESS} - W_{ESS,R}^{d-1}}{\eta_{ESS,in} \times T_v}. \quad (18)$$

where $P_{ESS,v}^3$ is the charging power of ESS for winter.

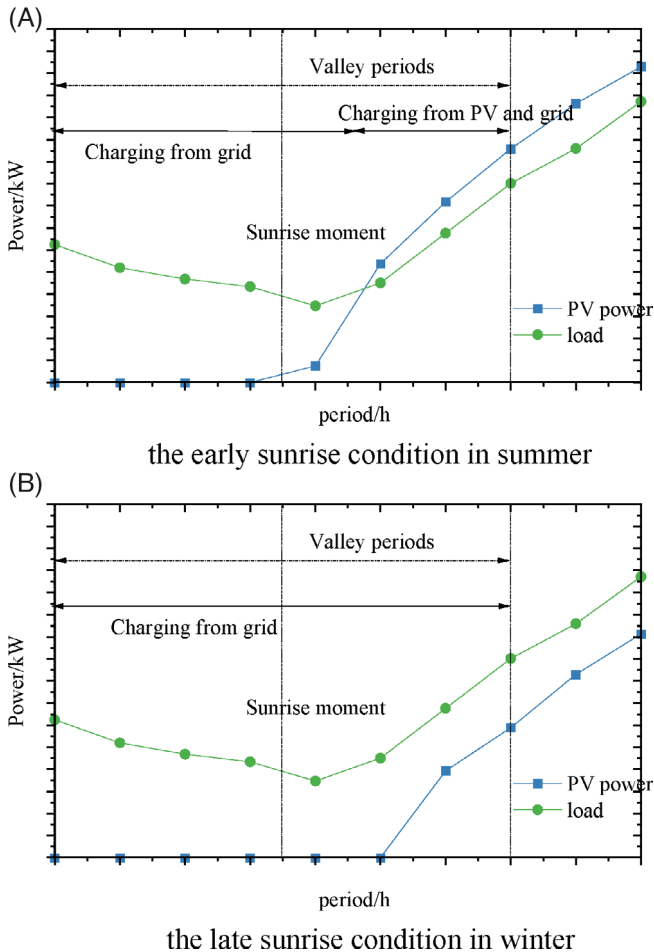


FIGURE 2 Charging strategy in the second stage: (A) for the early sunrise in summer and (B) for the late sunrise in winter

4.2 | Energy management strategy in the second stage

The second stage is the other time periods of the TOU, which is the PV power management and ESS and EV charging and discharging power management stage. Among them, PV management strategy is to sell part of the remaining PV power generated in each time period and store the other part; ESS and EV management strategy is to optimize the charging and discharging power of the battery in each time period according to the customer load and PV output power. This stage optimizes the PV consumption rate to achieve the peak regulation purpose while maximizing user benefits.

The solution of the second-stage model is to combine the results of the first-stage energy optimization management and calculate the total household microgrid benefits for a complete dispatch day according to the two-stage plan, which is solved using particle swarm algorithm. The specific energy management strategy is expressed as follows.

4.2.1 | Objective function

The objective of optimal energy management of household microgrid is to establish the optimal management strategy of the system when collaborative participation is managed without changing the travel habits of users and using EV charging and discharging functions to provide backup energy storage for the ESS system. To effectively reduce the cost of electricity for users and minimize the total cost of microgrid system operation, the objective function can be expressed as follows:

$$\min C = \sum_{\tau=1}^{24} (\mu_{\text{ESS}} |Q_{\text{ESS},\tau}| + \mu_{\text{EV}} |Q_{\text{EV},\tau}| + C_{\text{EI},\tau}), \quad (19)$$

$$C_{\text{EI},\tau} = P_{p,\tau} (Q_{p,\tau} - Q_{s,\tau}), \quad (20)$$

where $C_{i,\tau}$ is the cost of interaction with the main grid in the time period of τ ; $P_{p,\tau}$ is the price of electricity in the time period of τ ; and $Q_{p,\tau}$, $Q_{s,\tau}$ is the amount of electricity purchased and sold by the microgrid to the main grid in the time period of τ , respectively.

4.2.2 | Constraints

1. The micro-grid power balance constrain

$$P_{\text{PV},\tau} + P_{\text{ESS},\tau} + P_{\text{EV},\tau} + P_{p,\tau} - P_{s,\tau} = P_{U,\tau}. \quad (21)$$

2. The battery energy storage state constraint of ESS and EV: Equation (9).
3. The battery charge/discharge rate constraint of ESS and EV: Equation (10).
4. EV travel constraints: Equation (13).
5. The micro-grid interactive power constraint (Internet access and power purchase cannot be carried out simultaneously):

$$Q_{p,\tau} \times Q_{s,\tau} = 0 \quad (22)$$

4.2.3 | The solving of model

Based on the results of the first stage of energy optimization management, the model is solved by an improved particle swarm algorithm with the following steps:

1. Initialization data input. Input deterministic values of system PV, ESS, EV basic parameters, load demand, etc., energy management results of the first stage, and algorithm control parameters.
2. Population initialization. Generate the initial population with all particles satisfying the constraints under the constraint satisfaction test, and initialize the population position and velocity.
3. Solve the population target value as the fitness at this time, and judge the individual extremes and global extremes based on the fitness merit.
4. Update the particle positions and velocities. Perform iterative calculations to generate a new generation of particle populations, with position and velocity updated as follows:

$$v_{\rho}^{i+1} = v_{\rho}^i + l_1 r_1 (P_{\rho\text{OT}}^i - L_{\rho}^i) + l_2 r_2 (g_{\rho\text{OT}}^i - L_{\rho}^i), \quad (23)$$

$$L_{\rho}^{i+1} = L_{\rho}^i + v_{\rho}^{i+1}, \quad (24)$$

where v_{ρ}^i and L_{ρ}^i are the velocity and the position of the particle in the iteration of i ; l_1 and l_2 are the learning factors; r_1 and r_2 are the random numbers between $[0,1]$; $P_{\rho\text{OT}}^i$ and $g_{\rho\text{OT}}^i$ are the individual extremes and the global extremes of the population of the particle ρ in the iteration of i , respectively.

5. Iterate according to the updated particle swarm position and velocity to obtain the global optimal value.
6. Repeat (4) to (5) to search for the optimal solution until the maximum number of iterations is reached, and output the optimal solution and the

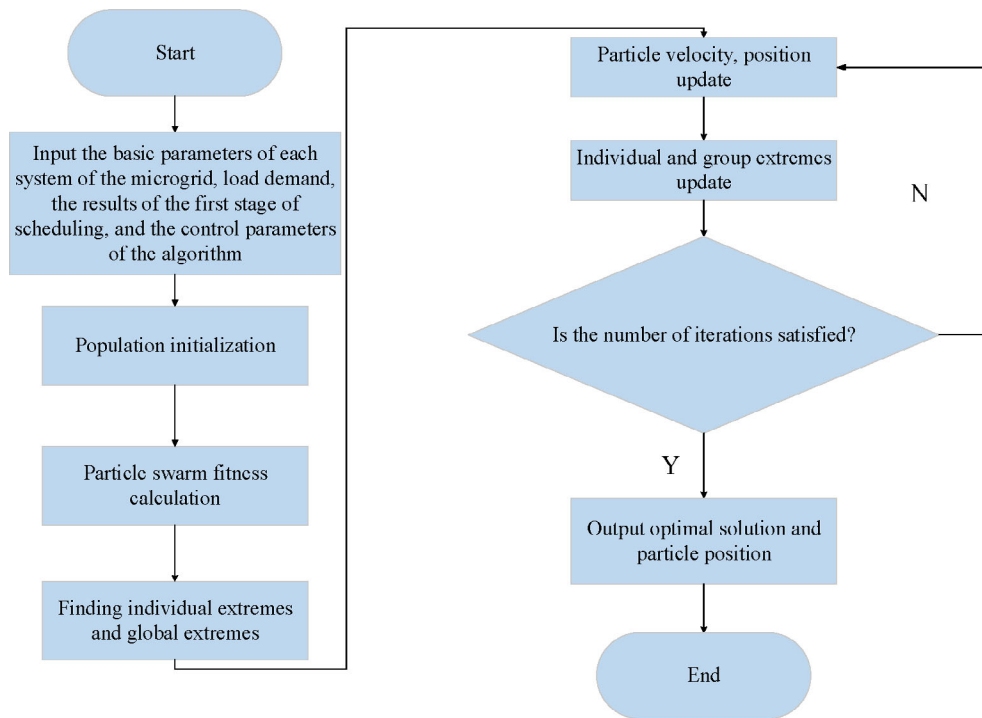


FIGURE 3 Algorithmic process in the second stage

TABLE 2 Peak and valley time-of-use electricity price

	Peak	Flat	Valley
Period/h	10:00-12:00, 18:00-24:00	8:00-10:00, 12:00-18:00	0:00-8:00
Electricity price (Yuan/kWh)	0.9323	0.5883	0.2433

TABLE 3 Technical parameters of EV

Parameter	Value	Unit
Power consumption per kilometer	0.13	kWh/km
Battery capacity	26	kWh
SOC upper limit	90	%
SOC lower limit	20	%
Charging/discharging efficiency	95	%

corresponding particle positions at this time. The flow chart of particle swarm algorithm can be represented as follows (Figure 3).

5 | RESULTS AND DISCUSSION

5.1 | Basic data

The rationality of the proposed model is verified based on a community dominated by household users, with typical

daily load demand data shown in Table 2 and time-of-use tariff shown in Table 3. The system parameters are set as follows: the total capacity of household PV is 0.6 MW, the ESS uses lithium iron phosphate batteries with long cycle life, the recommended capacity is 500 kWh, the charge/discharge efficiency is 95%, and the loss coefficient of energy storage batteries is 0.2. Fifty EVs can be called by the micro-grid, and the technical parameters of EVs are shown in Table 4.

Meanwhile, to study the influence of different weather and seasons on the optimization results, three typical scenarios are selected in this paper based on the PV output curves under different weather and seasonal conditions. They are summer sunny day (scenario 1), summer rainy day (scenario 2) and winter sunny day (scenario 3), and the PV output curves under each scenario are shown in Figure 4, assuming constant user load.

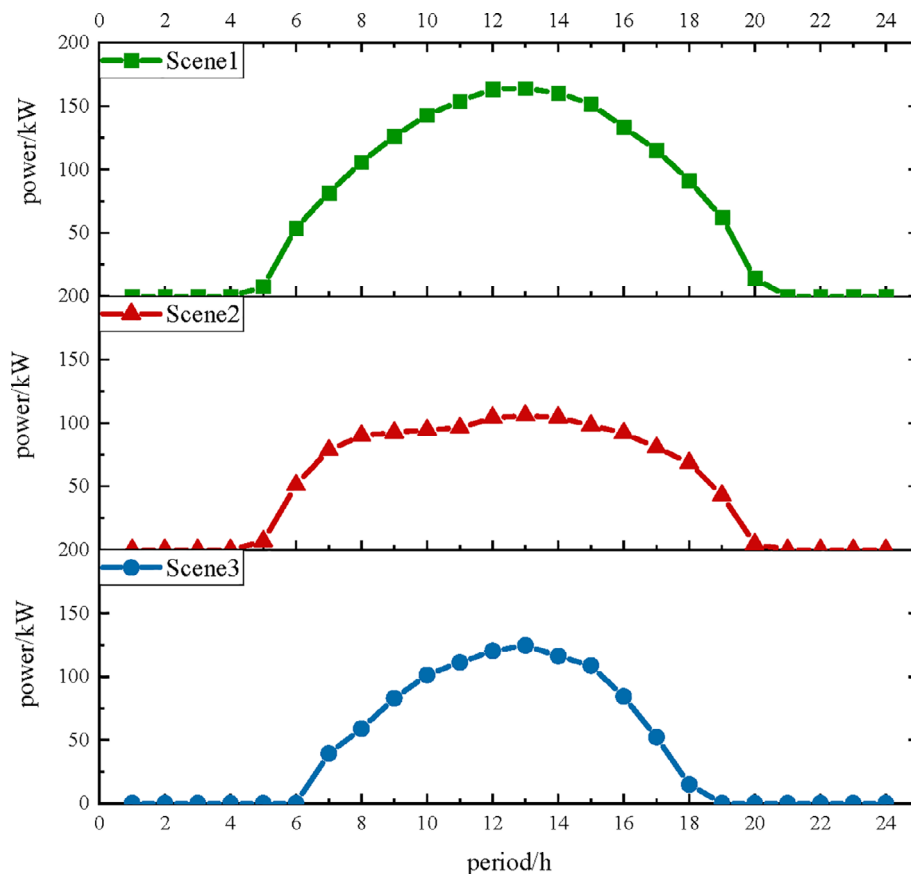
5.2 | Result analysis

5.2.1 | EV travel results

The distribution of EVs at different times of the day is calculated according to the Markov chain travel model, as shown in Figure 5. Figure 5 depicts the regional distribution results of 50 EVs under the premise of satisfying the random behavior, where the curves represent the number of vehicles in different regions at different time periods.

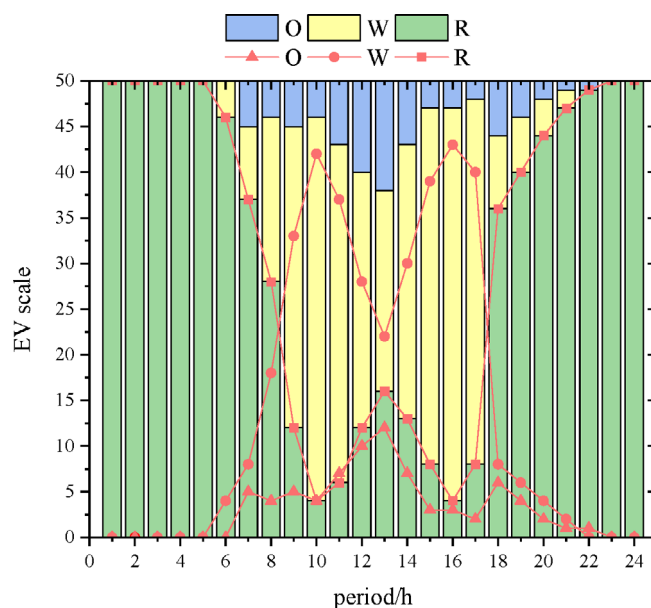
TABLE 4 Household microgrid operating costs in different scenes

Scene	Operating cost of ESS	Operating cost of EV	Electricity purchase cost	Electricity sales revenue	Total cost
1	186.49	364.00	353.63	570.08	334.04
2	147.57	364.00	358.99	150.45	720.12
3	154.34	364.00	390.27	90.75	817.87

**FIGURE 4** PV force curves in different scenes

According to the regional distribution results, it can be seen that the number of EVs in residential areas decreases significantly in the 7:00 to 9:00 time period and increases substantially in the 17:00 to 20:00 time period, which better fits the probability density curves of daily travel moments and daily return moments. Meanwhile, EVs are all parked in residential areas during 0:00 to 6:00 and 22:00 to 24:00 hours, and the number of EVs in residential areas during 6:00 to 22:00 hours is within the interval [4-49], so that the data in Equation 13 can be updated according to this calculation result.

The results of statistical trips show that the probability of EVs parked in residential areas is about 63%, and the proportion of time when all 50 EVs are parked in residential areas is about 30%, which is a relatively high proportion. Since EVs parked in residential areas can perform energy exchange, reduce the load fluctuation of home microgrid and reduce the cost of home microgrid

**FIGURE 5** EV regional distribution in different time periods

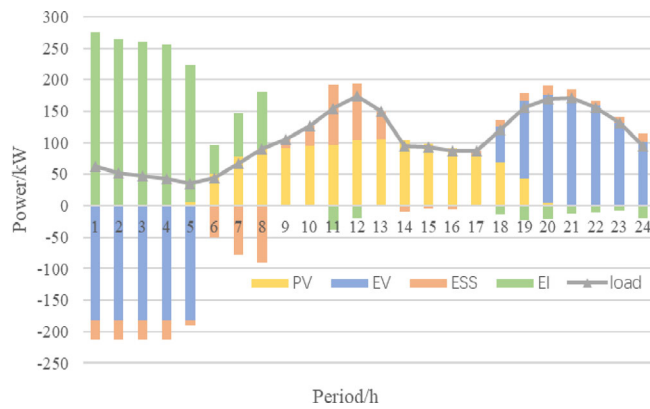


FIGURE 6 The results of the scheduling optimization in scenario 1

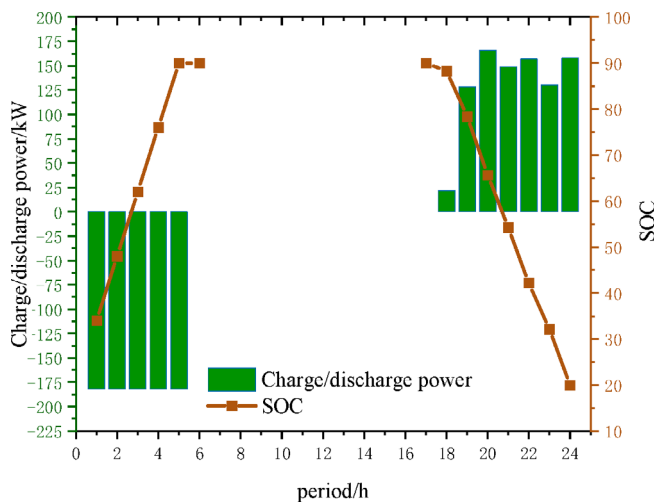


FIGURE 7 Charge/discharge power and SOC of EV in scenario 1

operation, and fully parked vehicles can significantly improve the efficiency of energy exchange, this home microgrid EV has the condition of energy optimization management.

5.2.2 | Optimization operation analysis

Set the particle swarm size as 80, the learning factor as 2, the maximum number of iterations as 200, and run the two-stage energy management model of home microgrid, assuming that the starting moment ESS and EV charge state are at the lowest point, that is, 0.2, taking scenario 1 as an example, the results of energy optimization management under TOU are as follows.

As shown in Figure 6, there is no sunlight during 0:00 to 4:00 hours, and it is in the valley hour of TOU, with low electricity price and low customer load level, so the

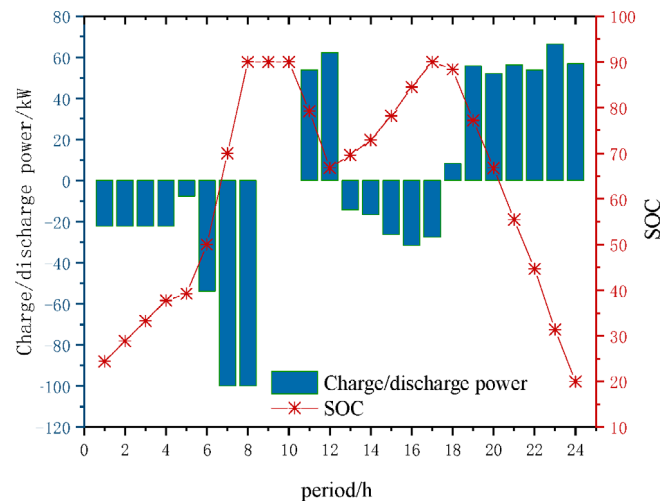


FIGURE 8 Charge/discharge power and SOC of ESS in scenario 1

load demand is met by purchasing electricity and making full use of low electricity price to charge EV; the sun starts to rise after 4:30, but the PV output power is not enough to meet the load demand at this time, so it is still necessary to purchase electricity from the grid; during 5:00 to 8:00 hours, although the light intensity is sufficient to support PV output is sufficient to cover the load demand, but still in the valley time, so continue to purchase power, using low electricity prices and PV residual power for ESS charging; 8:00 to 12:00 time, for the time-sharing tariff non-valley time, the load is growing, due to the summer sunny days, PV output can meet the growing demand for electricity, while collaborating with ESS to sell excess power, for the main network peak shaving at the same time to earn revenue. The load starts to drop between 12:00 and 18:00, and the PV output will be stored or sold to the grid on the basis of meeting the load demand; between 18:00 and 24:00, the load continues to increase, and the PV output gradually decreases to 0. EV starts to enter the grid in large quantities and gradually replaces the PV supply demand with ESS, and sends all the remaining power to the grid to relieve the main grid supply pressure. The EVs will gradually replace the PV supply demand and send all the remaining power to the grid to relieve the main grid supply pressure and earn the peak-to-valley price difference.

The charging and discharging power and charge states of ESS and EV during the optimization process are shown in Figures 7 and 8. As can be seen from the figures, ESS and EV operate according to the first-stage energy management strategy during the 0:00 to 8:00 time period, and the SOC is raised from a minimum of 0.2 to a maximum of 0.9. During the second-stage operation, during the 8:00 to 18:00 time period, ESS discharges during peak hours, and the SOC drops to a minimum of 0.67,

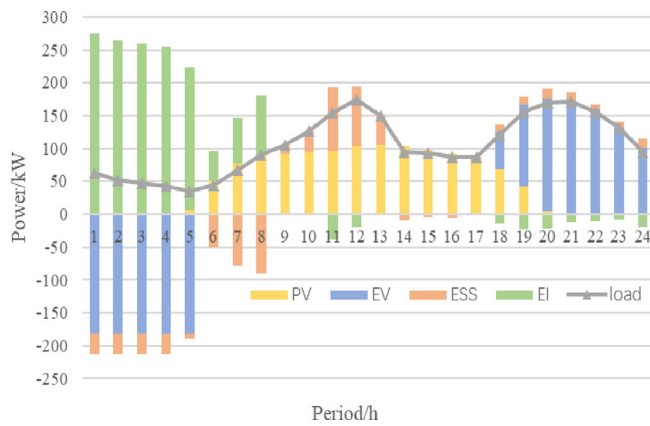


FIGURE 9 The results of the scheduling optimization in scenario 2

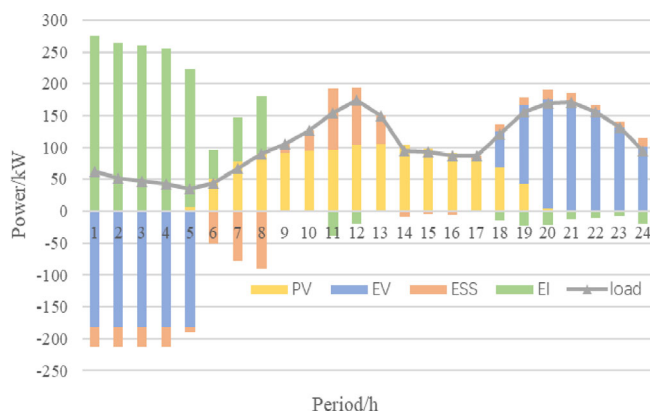


FIGURE 10 The results of the scheduling optimization in scenario 3

charges during weekdays, and continues to rise back to the 0.9 level; EV, because it only charges during the early morning valley hours. During the 18:00 to 24:00 hours, EVs enter the network in large quantities, and the SOC drops to 0.89 due to the drop in travel load, but the drop is not significant compared with the total power, after which they are discharged together with ESS to meet the internal load of the household micro-grid while supplying the main network.

Scenarios 2 and 3 are then optimized separately, and the optimization management results are shown in Figures 9 and 10. Combining with Figure 6, it can be seen that the sunrise time of scenarios 1 and 2 are located between 4 and 5 o'clock, and scenario 3 is located between 6 and 7 o'clock. Therefore, the optimization results of scenarios 1 and 2 in the first stage are similar, with the PV and the grid charging the ESS at the same time after sunrise, while the ESS power in scenario 3 is always purchased from the grid. In addition, the light intensity of scenarios 2 and 3 is lower than that of scenario 1, which makes the electric energy released by EV

and ESS in the second-stage optimization management process more for satisfying their own loads, compared with 660.24 kWh in scenario 1, and the electric energy sold to the grid at peak hours is significantly reduced, only 166.62 and 98.13 kWh, respectively. Scenarios 2 and 3 have the same specific optimization management strategies as scenario 1 in the two stages, and will not be described. Scenarios 2 and 3 are consistent with scenario 1 and will not be repeated.

5.2.3 | Cost-benefit analysis

The cost of each system of the home micronetwork is calculated for three different scenarios, and the summary results are shown in Table 5. It can be seen that the EV loss cost is the same for the three scenarios, and the ESS loss cost decreases. This is because the transportation property of EV makes EV only participate in charging and discharging once during the whole optimization management process, and EVs in all three scenarios are fully charged and discharged; while ESS is more flexible and participates in scheduling most frequently in scenario 1, followed by scenario 3 and least in scenario 2.

In addition, after optimization, the three scenarios can get \$570.08, \$150.45, and \$90.75 daily electricity sales revenue through peak-valley price difference, and the total daily cost of household micro-grid is \$334.04, \$720.12, and \$817.87, respectively, and the daily cost of household micro-grid electricity purchase without two-stage optimization management is \$1747.06. It can be seen that the use of two-stage energy management strategy in all three scenarios can reduce customers' electricity costs to some extent and improve the system economic efficiency.

5.2.4 | Sensitivity analysis

Based on the energy management results, it can be seen that EV battery capacity has a significant impact on the optimization results and system cost, so a sensitivity analysis is conducted on the number of dispatchable EVs to study the impact of EV battery capacity changes on system cost. The sensitivity of system cost to the number of dispatchable EVs under three scenarios is shown in Figure 11, taking the 50 EVs set in the example of this paper as the benchmark and varying in a range of 50%.

As can be seen from Figure 11, in scenario 1, the system cost decreases with the increase in battery capacity until the EV battery capacity reaches a certain level. However, after the battery capacity reaches a certain level, the system cost starts to increase with the increase of battery

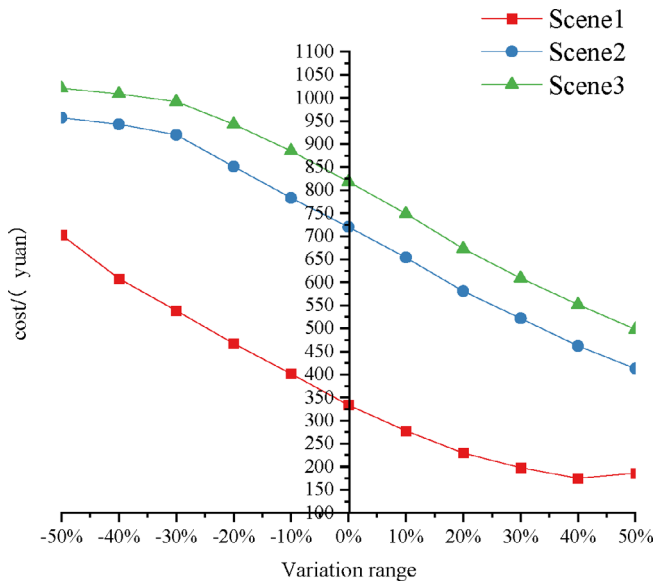


FIGURE 11 Sensitivity analysis diagram of EV quantity

capacity. The reason for this is that when the EV battery capacity reaches a certain level, the spread gain obtained by EV transferring power starts to be smaller than the loss cost, and there exists the optimal EV battery capacity that minimizes the system cost. In scenarios 2 and 3, the system cost always decreases with increasing battery capacity when the battery capacity varies within the 50% range. This is because under the lighting conditions of scenarios 2 and 3, even if the current EV battery capacity is increased by 50%, it still does not reach the optimal capacity in the respective scenarios. Therefore, in the next step of the study, the optimal EV battery capacity under different scenarios needs to be considered.

6 | CONCLUSION

This paper proposes a two-stage energy management strategy for household microgrid considering seasonal weather changes under the time-sharing tariff mechanism, and solves the optimal operation of each system using particle swarm algorithm, and finally selects three typical scenarios according to the PV output characteristics under different conditions, taking a community with mainly household customers as an example to verify the effectiveness of the proposed model. The following main conclusions are drawn from the study:

1. The proposed two-stage optimal energy management strategy for household microgrid can significantly improve the economic efficiency using time-of-use tariff, overcome the influence of different seasons and weather

on light intensity, and reduce the cost of electricity consumption for customers in all three typical scenarios.

2. The optimal management results obtained by solving with the particle swarm algorithm show that the two-stage energy management strategy can fully utilize the ESS and EV battery capacity, thus effectively improving the system economics.
3. The introduction of ESS and EV in household customers can not only earn price difference revenue, but also supply the grid load demand at the peak of electricity consumption, cut the peaks and fill the valleys, and realize the mutual benefit of household customers and power supply enterprises.

NOMENCLATURE

Acronyms

R	residential areas
W	work areas
O	other areas
H	power temperature coefficient
EV	electric vehicles
PV	photovoltaic
ESS	energy storage system
EI	energy interaction

Variables

$P_{PV,\tau}$	output power of household PV in the time period of τ
$SOC_{ESS,\tau}$	charge state of the energy storage battery in the time period of τ
$Q_{ESS,\tau}$	charge and discharge power of the energy storage system in the time period of τ
C_{ESS}	depreciation loss cost generated by battery energy storage's charging and discharging
$SOC_{EV,0}$	charge state of the EV at the time of grid entry
$Q_{EV,D}$	discharge of the EV between the time of travel and the time of return
$P_{EV,v}$	the first-stage EV charging power
$W_{EV,R}^{d-1}$	remaining EV battery capacity on the previous dispatch date
$P_{ESS,v}^1$	ESS charging power before sunrise in summer
$W_{ESS,R}^{d-1}$	remaining capacity of the ESS battery on the previous scheduling day
$P_{ESS,v}^2$	ESS charging power after sunrise in summer
$P_{U,\tau}$	power used by customers in the time period of τ
$P_{ESS,v}^3$	charging power of ESS for winter
$Q_{p,\tau}$	amount of electricity purchased by the microgrid to the main grid in the time period of τ
$Q_{s,\tau}$	amount of electricity sold by the microgrid to the main grid in the time period of τ

Parameters

T_b	start traveling time of the day
μ_b	mean of the travel time sample data
σ_b	variance of the travel time sample data
T_e	return time of the day
μ_e	mean of the return time sample data
σ_e	variance of the return time sample data
S_{EV}	daily mileage
μ_s	mean of the daily mileage sample data
σ_s	variance of the daily mileage sample data
$P_{ij,\tau}$	transfer probability from state A_i to state A_j in the time period of τ
P_{STC}	maximum test power at STC
G_{STC}	light intensity at STC
$G_{C,\tau}$	solar radiation intensity in the time period of τ
$T_{C,\tau}$	working temperature of PV panel in the time period of τ
$\eta_{ESS,in}$	charge efficiency of the energy storage battery
$\eta_{ESS,out}$	discharge efficiency of the energy storage battery
W_{ESS}	capacity of the energy storage battery
W_λ	power consumption per kilometer of the EV
W_{EV}	EV battery capacity
$D_{EV,\tau}$	the number of dispatchable EVs in the time period of τ
τ_T	EV travel time
SOC_{min}	lower limit of the state of charge
SOC_{max}	upper limit of the state of charge
μ_{ESS}	battery loss coefficient
τ_r	sunrise moment in summer
τ_v^{end}	end moment of the valley period
T_v	duration of the valley period
$C_{i,\tau}$	cost of interaction with the main grid in the time period of τ
$P_{p,\tau}$	price of electricity in the time period of τ
v_ρ^i	velocity of the particle in the iteration of i
L_ρ^i	position of the particle in the iteration of i
l_1	learning factors
r_1	random numbers between $[0,1]$
$P_{\rho OT}^i$	individual extremes of the population of the particle ρ in the iteration of i
$g_{\rho OT}^i$	global extremes of the population of the particle ρ in the iteration of i

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